Qoldau

AI Student Request Routing & Support System

Sprint 4 Review

User stories US17–US22 · what changed · live screenshots of the website

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SDU University · Sprint 4 deadline 06.11.2026

The product in one minute

Students write in their own words; the system finds the right office and tracks the request

1
Student writes
free text in KZ / RU / EN

2
AI classifies
10 categories + urgency

3
Routed
to the office & least-busy agent

4
Tracked
workflow, SLA timers, chat

5
Measured
analytics, CSAT, reports

Roles

Student · Agent · Unit head · Admin

Roadmap

Sprint 1
Core portal UI
US1–US4

Sprint 2
AI core: classify, priority, routing
US5–US10

Sprint 3
Tracking, SLA, analytics, admin, chat
US11–US16

Sprint 4
Export, retraining, smart suggestions, KB
US17–US22 · today

Sprint 5
Security, E2E tests, release
next

04

Sprint 4

Intelligent features: export, retraining, smart suggestions, knowledge base

Deadline 06.11.2026 · US17–US22

Sprint 4 backlog

Iteration 4 · from the S4 Backlog sheet

#	User story	Role	Points	Priority	Owner	Status
US17	**Report Export (Excel / PDF)**	Admin / Unit Head	5	LC & HV	Yessenbay Samatuly	Done
US18	**AI Retraining & Model Management**	Admin / System	8	HC & HV	Zhanibek Mukatay	Done
US19	**Smart Reply Suggestions for Agents**	Agent	6	HC & HV	Berik Zhumagaliyev	Done
US20	**Similar-Request & FAQ Suggestions**	Student / System	5	LC & HV	Alisher Beibituly	Done
US21	**Knowledge Base & FAQ Management**	Admin / Head / Agent	5	LC & HV	Madi Kanabekov	Done
US22	**Sprint 4 Scrum & Release Readiness**	Scrum Master	3	LC & HV	Madi Kanabekov (SM)	Done

Total: 32 story points · 6 stories · all Done

Build order: US21 Knowledge base first, because US19 and US20 read from it; US18 uses the agent corrections collected since Sprint 2 (US9). Goal: answers come faster, fewer requests are needed, and the AI keeps learning.

US21 Madi Kanabekov · 5 story points

Knowledge base & FAQ management

USER STORY

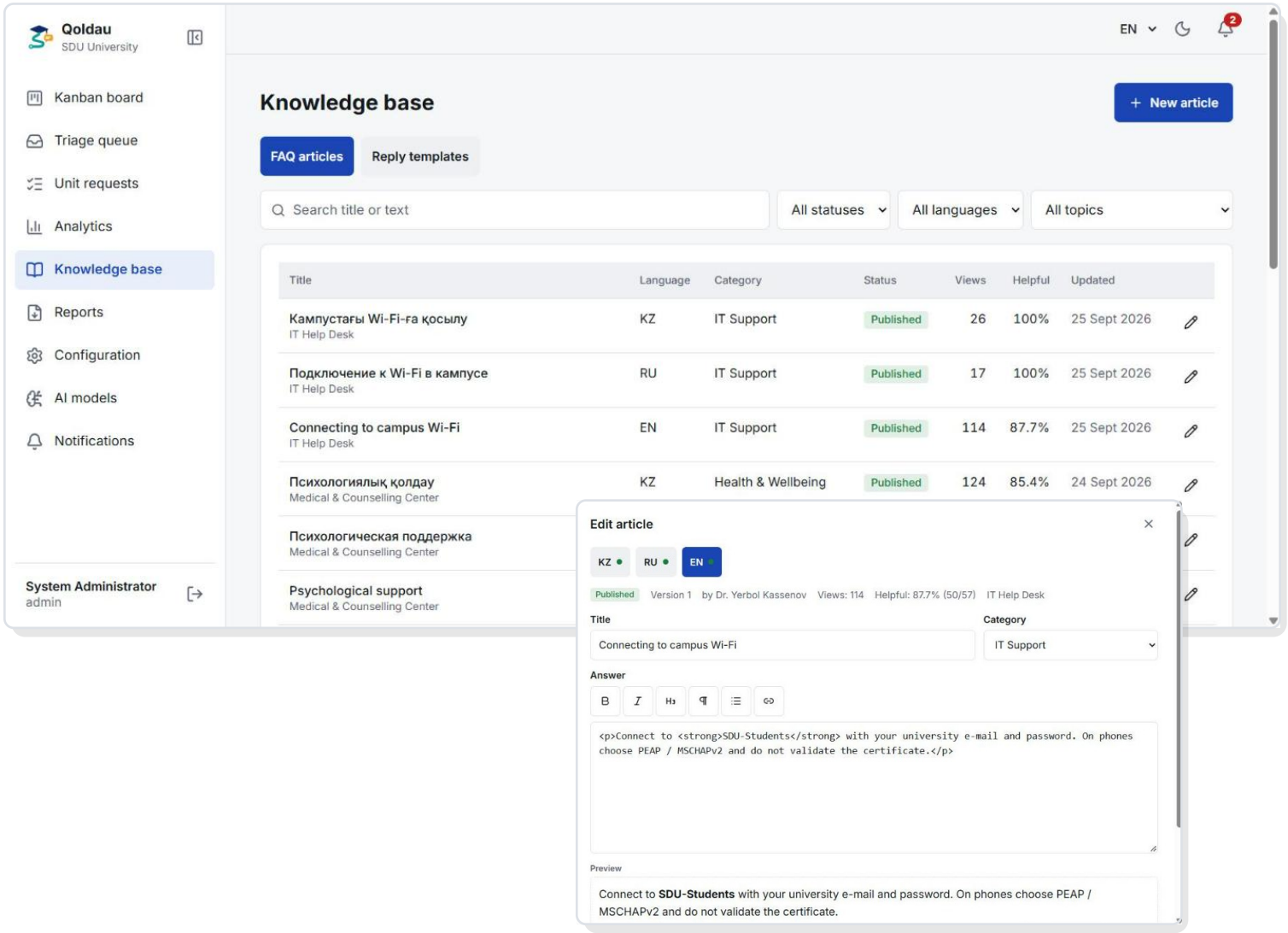
As an administrator or unit head, I want to create and maintain FAQ articles and reply templates per category and language so self-service and smart replies stay accurate.

ACCEPTANCE (GIVEN / WHEN / THEN)

- Registrar head creates "Ordering a transcript" in KZ / RU / EN and publishes: status Published, version 1, and within 1 min students get it as a suggestion.
- An agent clicks "Propose FAQ": a Draft is created, the head is notified, and the agent cannot publish (403).

WHAT WE BUILT

- 60 articles (20 topics × 3 languages) and 15 reply templates.
- Editor with language tabs, formatting and live preview; every edit saved as a version.
- Views, helpful %, usage shown per article.



US20 Alisher Beibituly · 5 story points

FAQ & similar-request suggestions for students

USER STORY

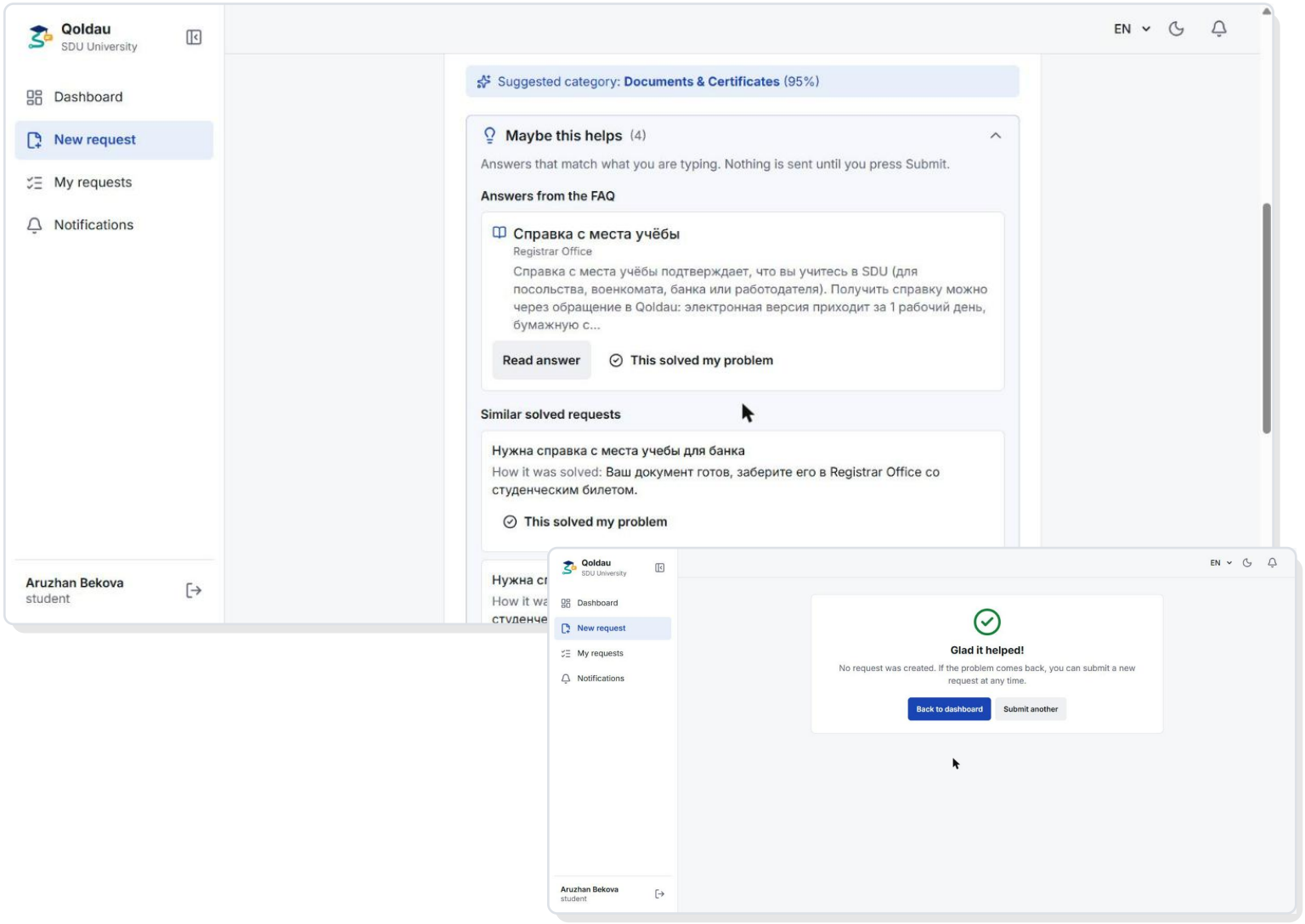
As a student, I want to see relevant FAQ answers and similar resolved requests while I type my request so I can solve simple issues myself without waiting.

ACCEPTANCE (GIVEN / WHEN / THEN)

- Typing «Как получить справку с места учебы» shows the "Enrollment certificate" FAQ (Registrar Office) in the top 3 within 800 ms; it opens inline.
- "This solved my problem" creates no request, records a deflection and raises the self-service rate.

WHAT WE BUILT

- Panel appears after 15 characters; works across Kazakh, Russian and English.
- Solved cases are anonymised (no names, e-mails, phones, IDs).
- Never blocks submitting; what was shown is saved with the request.



US19 Berik Zhumagaliyev · 6 story points

Smart reply suggestions for agents

USER STORY

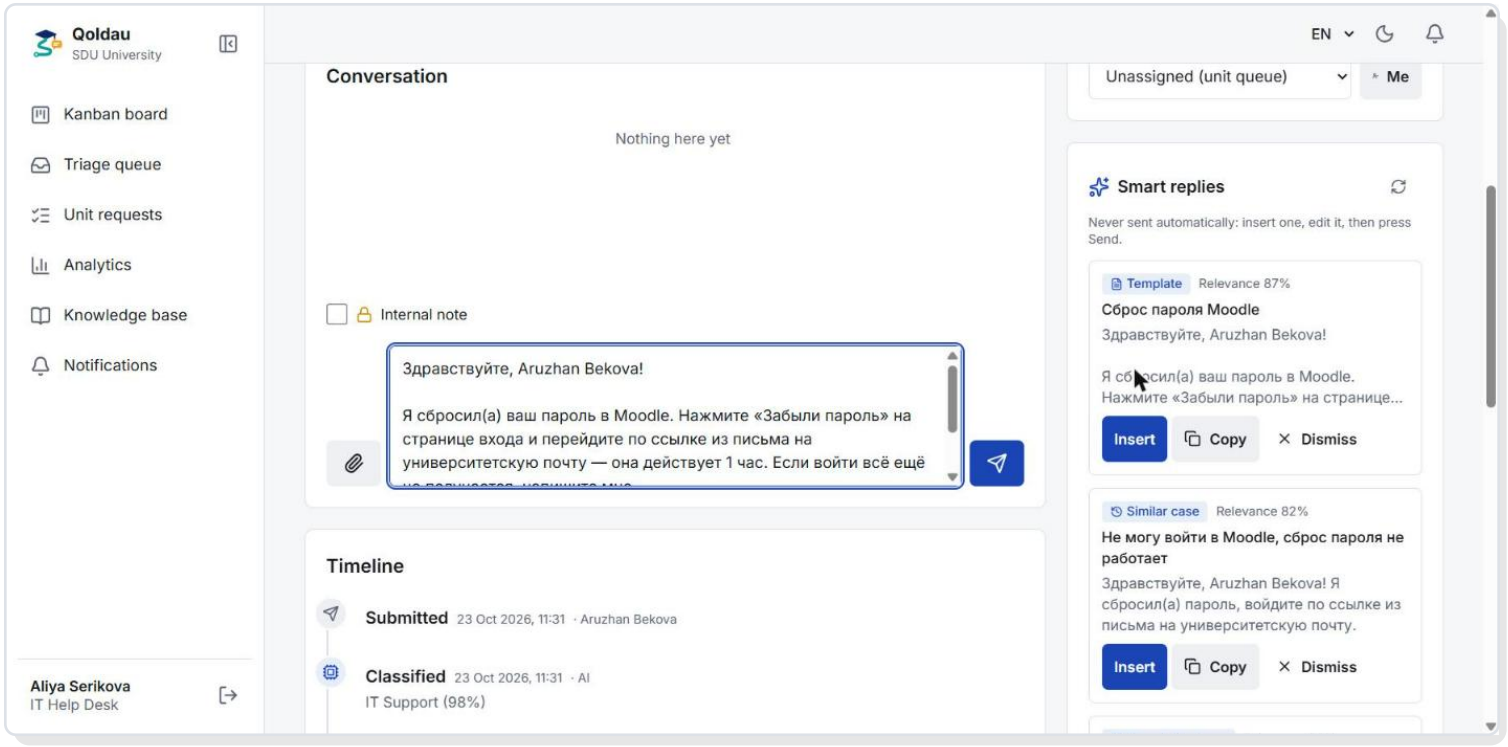
As an agent, I want suggested replies based on the request text, similar resolved cases and knowledge-base articles so I can answer students faster and consistently.

ACCEPTANCE (GIVEN / WHEN / THEN)

- IT agent opens “Не могу войти в Moodle” (RU): 3 Russian suggestions in under 1.5 s with source and relevance; the Moodle template is #1.
- “Insert” fills {student_name}, {agent_name}… into the reply box and does not send.
- Other students' data is masked, internal notes are never used; used / edited / ignored is logged.

WHAT WE BUILT

- Three sources: reply templates, similar solved cases, FAQ articles.
- Answers in the request's language in about 20 ms.
- Smart-reply usage KPI on the dashboard.



US18 Zhanibek Mukatay · 8 story points

AI retraining & model management

USER STORY

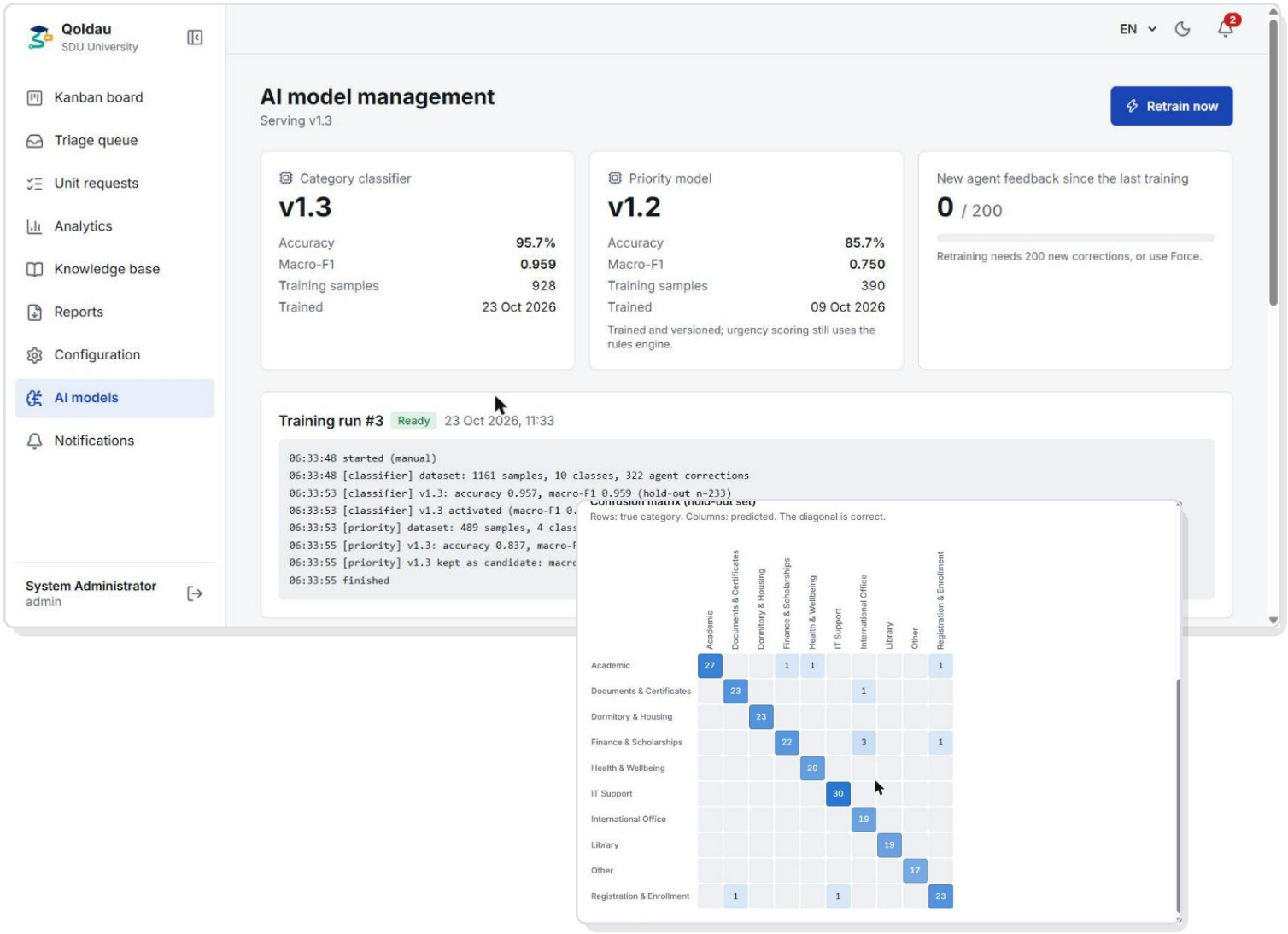
As an administrator, I want the AI model to be retrained from agents' corrections and to manage model versions so classification accuracy keeps improving over time.

ACCEPTANCE (GIVEN / WHEN / THEN)

- 250+ new corrections, active v1.2: “Retrain” trains, evaluates and registers v1.3 with accuracy, macro-F1 and confusion matrix; v1.3 becomes active because its F1 is not lower; the switch is logged.
- “Roll back” to v1.2 works within a minute, without downtime.

WHAT WE BUILT

- Live demo: macro-F1 0.912 (v1.2) → 0.959 (v1.3), promoted automatically.
- A weaker model stays a candidate (priority model: 0.728 < 0.750).
- Per-category F1, confusion matrix, version history.



US17 Yessenbay Samatuly · 5 story points

Report export (Excel / PDF)

USER STORY

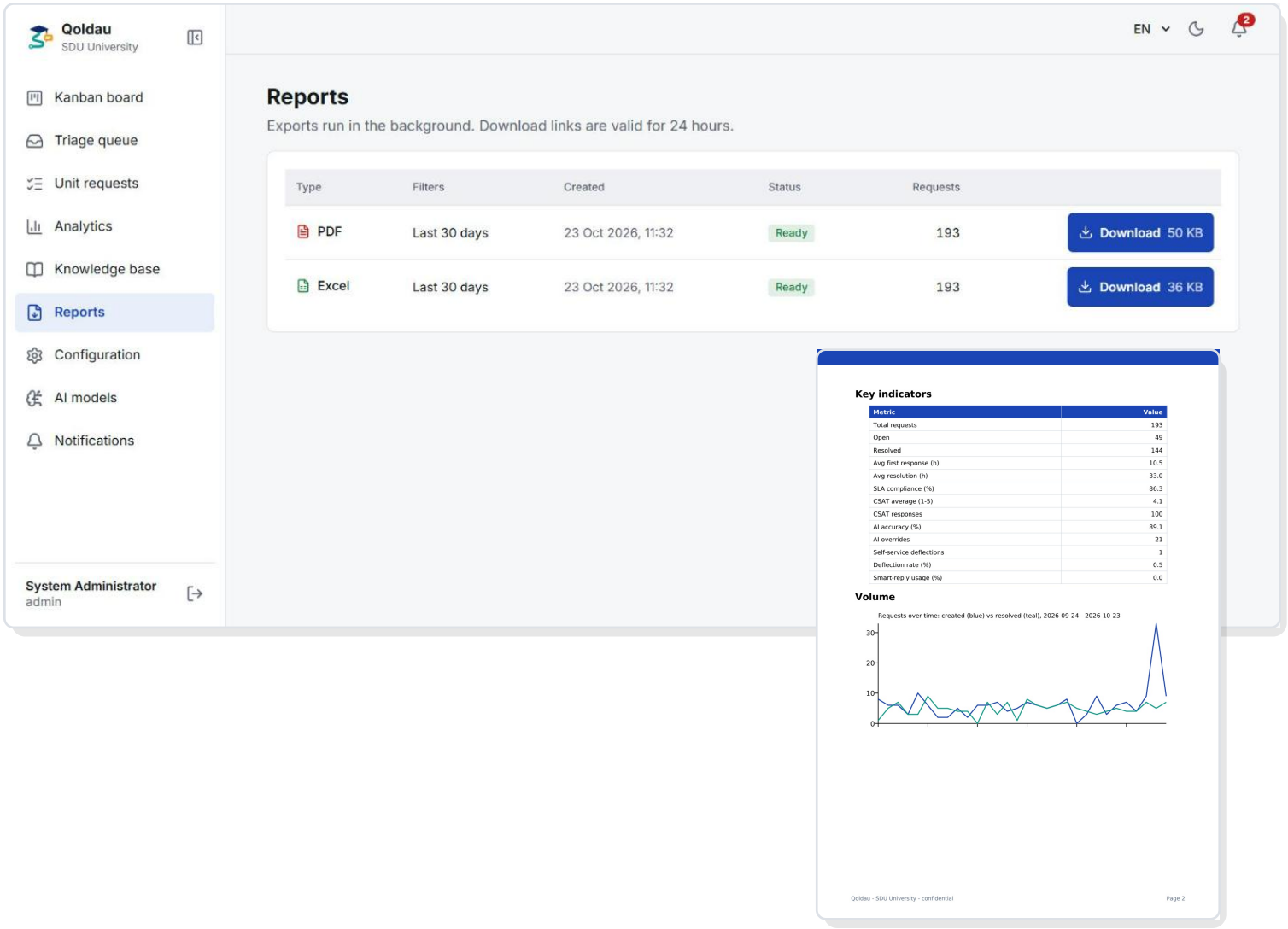
As a university administrator, I want to export analytics and filtered request lists to Excel and PDF so I can share reports with management without manual work.

ACCEPTANCE (GIVEN / WHEN / THEN)

- Dashboard filtered by period and unit → Export → Excel: within 30 s a file with sheets Summary, By Category, By Unit, SLA, AI Accuracy, Raw Requests; totals equal the dashboard; logged in the audit log.
- Finance unit head's PDF contains only Finance requests; a student gets 403.

WHAT WE BUILT

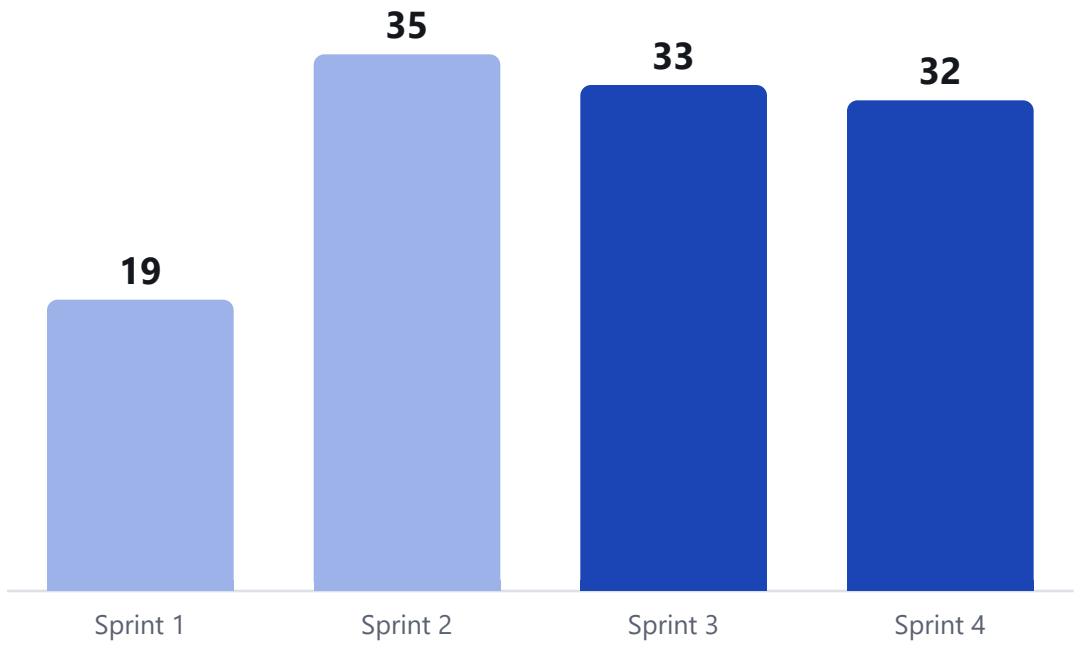
- Export menu on Analytics and Unit requests, using the current filters.
- Runs in the background → bell notification → Reports page, link valid 24 h.
- 10,000 rows in 2.5 s; student e-mails masked unless Admin.



Scrum management & quality

US16 (Sprint 3) and US22 (Sprint 4): ceremonies, velocity, tests

Velocity (story points completed)



Sprints 3 and 4 kept a steady pace (33 and 32 points) after the larger AI sprint.

How we worked

- Sprint Planning with the HC/LC × HV/LV matrix; daily stand-ups; Review + Retrospective
- Dependencies found at planning: US21 Knowledge base before US19/US20; US9 feedback data before US18
- Every story checked against its USxQATest Given/When/Then scenarios

Quality in numbers

58 automated tests, all passing (40 → 58 in Sprint 4)

12/12 Sprint 4 Given/When/Then scenarios automated

2.5 s 10,000-row Excel export (target 30 s)

~20 ms smart replies (target 1.5 s); FAQ suggestions < 50 ms (target 0.8 s)

What changed in the system

New parts added in Sprint 3 and Sprint 4 (blue = Sprint 4)

Website (Next.js)

Request timeline & search	S3
Kanban SLA badges	S3
Analytics dashboard	S3
Configuration panel	S3
Chat + rating	S3
“Maybe this helps” panel	S4
Smart replies panel	S4
Knowledge base	S4
AI models page	S4
Reports + Export menu	S4

API (FastAPI)

/requests/{id}/timeline	S3
/analytics/*	S3
/admin/*	S3
/messages, /csat	S3
/ai/similar, /ai/suggestions	S4
/admin/faq, /reply-templates	S4
/ai/retrain, /ai/models	S4
/export/excel, /export/pdf	S4

Services & AI

SLA scheduler + escalation	S3
Audit log	S3
WebSocket chat	S3
Embeddings + similarity search	S4
PII masking	S4
Retraining + model registry	S4
Excel / PDF report builder	S4
Background jobs	S4

Demo accounts

Password for every account: Qoldau123!

Student	student@qoldau.edu.kz	New request + FAQ suggestions, My requests, timeline, chat, rating
Agent · IT Help Desk	agent1.it@qoldau.edu.kz	Kanban board, ticket with Smart replies, Propose FAQ
Agent · Triage	agent1.triage@qoldau.edu.kz	Triage queue (low-confidence requests)
Unit head · Registrar	head.registrar@qoldau.edu.kz	Publish FAQ articles, unit analytics, export
Unit head · Library	head.library@qoldau.edu.kz	Analytics locked to own unit
Admin	admin@qoldau.edu.kz	Configuration, AI models, Reports, all analytics

Other units follow the same pattern: agent1-3.<unit>@qoldau.edu.kz and head.<unit>@qoldau.edu.kz (deans, registrar, finance, it, housing, library, medical, international, triage).

Next: Sprint 5

Release readiness

Security
full RBAC audit, rate limiting, OWASP Top-10 review

Testing
end-to-end tests (Playwright), coverage ≥ 70%

Performance
load tests, PostgreSQL + pgvector in production

Release
CI/CD, deployment, final demo

Thank you!

Questions?

Live demo: docs/SPRINT4_DEMO.md

make demo → fresh data in ~20 s